

LALITH KOUSHIK VANAM

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PROFESSIONAL SUMMARY

Full Stack & AI Software Engineer with experience building production-grade web platforms and agentic AI systems for the oil & gas pipeline and healthcare domains. Skilled in multi-agent AI orchestration over relational and graph data, full-stack development with ASP.NET Core and React, and secure, well-architected backend design. Proven track record translating complex operational requirements into maintainable software for safety-critical environments. Currently extending hands-on expertise into SCADA systems, industrial communications, and ICS cybersecurity.

TECHNICAL SKILLS

Backend & APIs	ASP.NET Core MVC, Web API, FastAPI, GraphQL, RESTful Services
AI & LLM Systems	Multi-Agent AI Systems, LLM Orchestration, RAG Pipelines, Azure OpenAI, NetworkX, Conversational AI, Model Tiering
Frontend & Viz	React.js, Vite, TypeScript, Plotly.js, JavaScript, HTML/CSS
Databases	PostgreSQL, SQL Server, MySQL, MSSQL, HeidiSQL
Architecture	Clean Architecture, Repository Pattern, Microservices, API-first Design
Industrial / SCADA	Ignition SCADA, SCADA Architecture, RTU/PLC Communications, Modbus & OPC-UA, HMI Design, Alarm Management, PI Historian, dataPARC, Flowcal
Standards	ANSI/ISA-62443, API 1164, API 1165, NIST SP 800-82, PHMSA
DevOps & Tooling	CI/CD Pipelines, Git, Docker, Cloudflare

PROFESSIONAL EXPERIENCE

MidstreamAI **Dec 2024 – June 2026**
Full Stack & AI Software Engineer *Houston, TX*

- Designed and owned a production-grade multi-agent AI analytics platform that lets operators query pipeline infrastructure in natural language, with a supervisor agent routing to specialized topology and operations agents, intelligent request planning, structured error recovery, result caching, and tiered model selection for reliability and cost efficiency.
- Designed and delivered a secure, enterprise pipeline leak-detection analytics platform (ASP.NET Core MVC, Plotly.js) featuring interactive dashboards, server-side aggregation, policy-based RBAC, hybrid EF Core / Dapper data access, and centralized logging with reusable components adopted across multiple applications.
- Collaborated with domain experts to translate operational requirements into production software, following structured change-management practices.

University of Houston-Clear Lake **Aug 2025 – May 2026**
Graduate Teaching Assistant *Houston, TX*

- Supported five graduate/undergraduate courses: Computer Security & Cryptography, Computer Game Programming, Concurrent Programming & Software Modelling, Network Protocols, and Design of Database Systems.
- Clarified course concepts and guided students through programming assignments and projects during office hours and lab sessions.
- Graded programming assignments, lab reports, and course projects against consistent evaluation standards; proctored quizzes and exams and upheld academic integrity.

LTIMindtree **Feb 2024 – Jun 2024**
Full Stack Developer *Hyderabad, India*

- Gained exposure to industrial seep detection and leak analysis systems in oil and gas pipeline monitoring environments, developing conceptual familiarity with SCADA-adjacent operational workflows.
- Learned about alarm analysis workflows and trend-based visualizations using dataPARC (pipeline SCADA historian) for operational data monitoring in pipeline environments.
- Gained foundational understanding of operational reliability, data accuracy, and system validation requirements in safety-critical pipeline applications.

SELECTED PROJECTS

Agentic-AI Analytics Platform
FastAPI • Multi-Agent Orchestration • PostgreSQL • NetworkX • React/TypeScript

- Built a microservices-based multi-agent AI platform enabling natural-language queries over pipeline infrastructure: a supervisor agent routes requests to specialized topology and operations agents and synthesizes their results into clear answers with full audit logging.
- Engineered an intelligent request-routing layer that classifies each query, plans the work upfront as dependency-aware tasks, and asks clarifying follow-up questions (with selectable options) when intent is ambiguous, avoiding wasted work and incorrect answers.
- Tuned the agent workflow for reliability and cost: structured error recovery separating recoverable from terminal failures, result caching, automatic retry with backoff for transient errors, and tiered model selection (lightweight models for routine steps, stronger models for synthesis).
- Implemented a topology agent reasoning over a graph model of the pipeline network and an operations agent querying time-series sensor data, both feeding a shared state object for unified, context-aware responses; added session-level conversation memory for reference resolution across turns, real-time streaming APIs, and automated Plotly.js visualizations.

Pipeline Leak Detection Insights Application
ASP.NET Core MVC • Plotly.js • MSSQL • EF Core • Dapper • Serilog

- Developed a production-ready analytics application for pipeline leak detection with interactive Plotly.js dashboards, repository-pattern data access, server-side aggregation controllers, and dynamic client-side filtering, designed for scalability and maintainability in an industrial environment.
- Implemented hybrid data access using EF Core for mapped domain entities and Dapper for high-performance raw SQL, balancing developer productivity with query performance; configured dependency injection with explicit lifetimes (transient DB connections, scoped services, controlled DbContext lifecycle) to reduce resource contention and leaks.
- Enforced policy-based, database-driven authorization with a custom requirement/handler for fine-grained screen-level RBAC; secured user flows with cookie-based authentication (configurable login/access-denied paths, session expiration) and a request filter that redirects on session expiry.
- Centralized structured logging with Serilog (configurable minimum level, daily rolling files) and added global exception handling that logs exceptions plus request/response details for faster root-cause analysis; built a test helper logger to capture in-memory logs for deterministic test assertions.
- Hardened deployment and API behavior via middleware and hosting tweaks (cookie policy, IIS synchronous IO, HTTPS redirection, static files), JSON serialization configured to ignore reference cycles with indented output, runtime Razor compilation for faster front-end iteration, and client-side integration of rich UI libraries (interactive tables, math rendering).

NasaFitness: Full-Stack Employee Wellness & Telehealth Platform
ASP.NET Core Web API • React • SQL Server • Azure OpenAI • GraphQL

- Built a HIPAA-aware employee wellness platform with three role-based portals (employee, employer, clinician) spanning health screening, fitness assessment, AI-generated wellness plans, and telehealth.
- Designed a multi-role JWT auth system with BCrypt password hashing, claims-based authorization, and 8 named permission policies derived from role at token issuance.
- Built a clinical intake pipeline with PAR-Q health screening, fitness exam capture (BMI, body-fat %, grip strength, flexibility), and physician medical-clearance routing via a fax/email gateway; implemented NASA-standard clinical calculations (3-site skinfold body-fat regression, lean body mass, gender-specific ideal weight) as a reusable service.
- Added global exception-handling middleware to mask sensitive error details from clients while preserving structured server-side logs; exposed a GraphQL DashboardQuery alongside REST for aggregated employee wellness metrics.
- Integrated Azure OpenAI (gpt-4o) to generate Personal Wellness Plans from patient context (health, nutrition, sleep, and stress inventories plus fitness exam results), with eligibility gating to require sufficient clinical data before generation.
- Integrated Microsoft Graph API (OAuth2 client-credentials) to fetch Microsoft Teams meeting transcripts, with a background TranscriptDiscoveryWorker that batches polling to avoid API throttling; integrated Acuity Scheduling (REST + webhooks) for telehealth booking.
- Built a fax gateway using RingCentral (multipart form + Bearer auth) with SMTP fallback to deliver clinically-formatted PAR-Q PDFs (QuestPDF, with preserved official wording and layout) to clearing physicians.

SCADA TRAINING (Self-Directed, Ongoing)

- **Hands-On Platform (Ignition SCADA):** Building a working pipeline-tank-farm SCADA application in Ignition, configured device simulators and a structured tag hierarchy, designed a Perspective HMI with live-bound indicators, gauges, control buttons, and historical trend charts, now implementing alarm configuration and management.
- **SCADA Architecture & Communications:** Field-instrument-to-historian control system architecture (PLCs/RTUs to SCADA server); Purdue Model and OT/IT separation; control-center redundancy; Modbus and OPC-UA protocols; communication mediums and failover.
- **HMI, Alarms & Historian:** HMI components and operator workflows; alarm priority levels and API 1165 display design; PI Historian tag configuration and time-series storage; Flowcal and custody-transfer concepts.
- **ICS Cybersecurity:** ANSI/ISA-62443, API 1164, and NIST SP 800-82 frameworks; OT/IT network segmentation and secure architecture for industrial control systems.

EDUCATION

University of Houston <i>Master of Science in Computer Science</i>	Aug 2024 – May 2026 <i>Houston, TX</i>
Keshav Memorial Institute of Technology <i>Bachelor of Technology in Computer Science</i>	2020 – 2024 <i>Hyderabad, India</i>